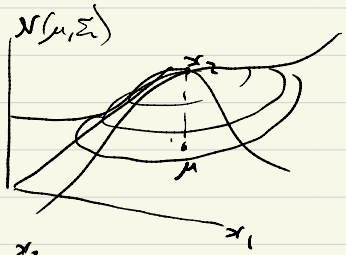
Lecture 5

- Multivariate Gaussian dist.
- Expectation, mean, covariance, moments
- Jointly Gaussian R.V.s

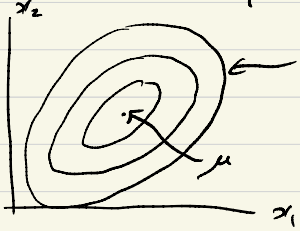
Recall the multivariate Gaussian

$$X \sim N(\mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

mean $\in \mathbb{R}^n$ ↑ covariance $\in \mathbb{R}^{n \times n}$

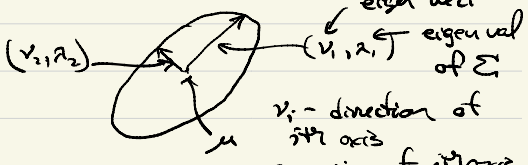


top down



single ellipse we

confidence ellipse:



Q: How to get eq. of confidence ellipse?

A: for level c:

confidence ellipse: $\{x \mid N(\mu, \Sigma) \geq c\}$

let's put in a more function form:

let's define normalization constant:

$$\eta := \frac{1}{\sqrt{(2\pi)^n |\Sigma|}}$$

$$N(\mu, \Sigma) = \eta e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} = c$$

$$-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu) = \log\left(\frac{c}{\eta}\right)$$

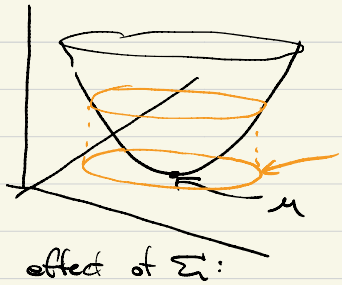
$$(x-\mu)^T \Sigma^{-1}(x-\mu) = \underbrace{2\log\left(\frac{\eta}{c}\right)}_{:= K(c)}$$

we see confidence ellipse boundary:

$$\{x \mid \underbrace{(x-\mu)^T \Sigma^{-1}(x-\mu)}_{\text{eq. of a quadratic "bowl"}} = K(c)\}$$

for Σ^{-1} symmetric pos. def.

visualize:



consider effect of Σ:

- $\Sigma \uparrow \Rightarrow \Sigma^{-1} \downarrow \Rightarrow$  $\Rightarrow$ high uncertainty

- $\Sigma \downarrow \Rightarrow \Sigma^{-1} \uparrow \Rightarrow$  $\Rightarrow$ low uncertainty

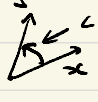
define | Information Matrix: $\Omega := \Sigma^{-1}$ |

notice: both Σ and Ω are s.p.d.

p.d. def:

$$x^T \Sigma x > 0 \quad \forall x \neq 0$$

consider Σ^{-1} define $y = \Sigma x$



so $x = \Sigma^{-1} y$

$$\underbrace{(\Sigma^{-1} y)^T}_{x^T}(\Sigma \Sigma^{-1} y) = y^T \Sigma^{-1} y$$

$$= x^T \Sigma x > 0 \quad \forall x \neq 0$$

so $y^T \Sigma^{-1} y > 0 \quad \forall y \neq 0$

$\Rightarrow \Sigma^{-1}$ p.d. □

Standard Normal R.V.

$\boxed{X_s \sim N_s(0, I)}$ so $\mu_s = 0$, $\Sigma_s = I$



contour "ellipses" are circles

- we can construct any n-D Gaussian R.V. X from X_s
- we can transform any n-D Gaussian X to X_s with coord. transform

specifically,

$\boxed{X = \Sigma^{1/2} X_s + \mu}$

$\boxed{\text{gives } X \sim N(\mu, \Sigma)}$

where $\Sigma^{1/2}$ is the matrix square root so $(\Sigma^{1/2})^T \Sigma^{1/2} = \Sigma$.

most common choice Cholesky square root we'll unpack this later.

and we see:

$\boxed{X_s = \Sigma^{-1/2}(X - \mu)}$

$\boxed{\text{gives } X_s \sim N(0, I)}$

Expectation, Mean, Covariance, Moments

mean: 1st moment $\in \mathbb{R}^n$

$\boxed{E[X] = \int_x x\, p(x)\, dx}$

we say "expected value of X"

we also call this the mean: $\boxed{\mu = E[X]}$

"average value of X"

covariance: 2nd moment $\in \mathbb{R}^{n \times n}$

$\boxed{E[(X-\mu)(X-\mu)^T] = \int_x (x-\mu)(x-\mu)^T p(x)\, dx}$

we give special symbol

$\boxed{\Sigma = \text{Cov}(X) = E[(X-\mu)(X-\mu)^T]}$

Notice Gaussians are entirely determined by (μ, Σ) so call 2nd order distributions (μ, Σ) 2nd order statistics

3rd moment $\in \mathbb{R}^{n \times n \times n}$ $\quad \left[E[(x_i - \mu_i)(x_j - \mu_j)(x_k - \mu_k)]\right]_{ijk}$

"skew"

4th moment $\in \mathbb{R}^{n \times n \times n \times n}$ $\quad \left[E[(x_i - \mu_i)(x_j - \mu_j)(x_k - \mu_k)(x_l - \mu_l)]\right]_{ijkl}$

"kurtosis"

For the Gaussian it happens to be true that:

$$\mu = \int_x x\, \eta\, e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}\, dx$$

and

$$\Sigma = \int_x (x-\mu)(x-\mu)^T \eta\, e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}\, dx$$

you will show in PSet 2.

Notice these moments are all special cases of expectation of a function of a R.V.

in general:

for a function $f(X)$

$\boxed{E[f(X)] = \int_x f(x)\, p(x)\, dx}$

E.g. if $f(X) = X \Rightarrow$ mean

if $f(X) = (X-\mu)(X-\mu)^T \Rightarrow$ cov.

$\vdots$

Consider a R.V. $Y \sim p(y)$ but Y arises as a function of R.V. $X \sim p(x)$ so $Y = f(X)$

we have:

$$\boxed{E_y[Y] = \int_y y\, p(y)\, dy = E_x[f(X)] = \int_x f(x)\, p(x)\, dx}$$

E.g. if we want to compute $E[Y]$ but we don't know $p(y)$ (or it's hard to compute). But we do know $f(X) = Y$ and we do know $p(x)$. Then we can still compute $E[Y]$

$\boxed{E_y[Y] = \int_x f(x)\, p(x)\, dx}$

without knowing $p(y)$!

Called LOTUS: Law of the Unconscious Statistician

E.g. Suppose Y is a sensor measurement and X is a state

Suppose we have a sensor model

$$Y = AX + b$$

affine model so $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$

we can compute $E[Y]$ without knowing $p(y)$!

$$E[Y] = \int_x (Ax + b) p(x) dx$$

$$= A \int_x x\, p(x) dx + b \int_x p(x) dx$$

($E[X]$ and 1)

$$\boxed{E[Y] = AE[X] + b}$$

or $\boxed{\mu_y = A\mu_x + b}$

also note $E[\cdot]$ is a linear operator

Empirical Moments

theoretical ($p(x)$ known)	empirical (N samples $X_i \sim p(x)$ i.i.d.)
$\mu = E[X] = \int_x x p(x) dx$	$\iff \quad \bar{\mu} = \frac{1}{N} \sum_{i=1}^{N} X_i$
$\Sigma = \int_x (x - \mu)(x - \mu)^T p(x) dx$	$\iff \quad \bar{\Sigma} = \frac{1}{N} \sum_{i=1}^{N} (X_i - \bar{\mu})(X_i - \bar{\mu})^T$
$\vdots$ any moment	
$E[f(X)] = \int_x f(x) p(x) dx$	$\iff \quad \approx \frac{1}{N} \sum_{i=1}^{N} f(X_i)$

Law of Large Numbers establishes that as $N \to \infty$ the empirical estimates approach the theoretical integrals.